Observation - Week 05

Topic: Full Stack Integration (MERN)

Name: _____ Roll No: _____ Date: _____

Aim

To integrate a React frontend with an Express and MongoDB backend into a working full stack application.

Concepts Covered

- Client-server architecture
- CORS and API base URLs
- React useState / useEffect with fetch
- Environment variables with dotenv

Observations

- Without the cors middleware the browser blocks requests from port 5173 to port 5000.
- useEffect with an empty dependency array loads data once on mount.
- The Network tab shows each request's method, status and timing.
- Secrets such as MONGO_URI belong in .env, which is excluded by .gitignore.

Viva Questions

1 What is CORS?

2 What does useEffect do?

3 What is the MERN stack?

4 How do you structure a client/server repo?

Reference Files

- src/server/server.js
- src/client/src/App.jsx
- screenshots/frontend-ui-output.png
- screenshots/network-tab-api-call.png